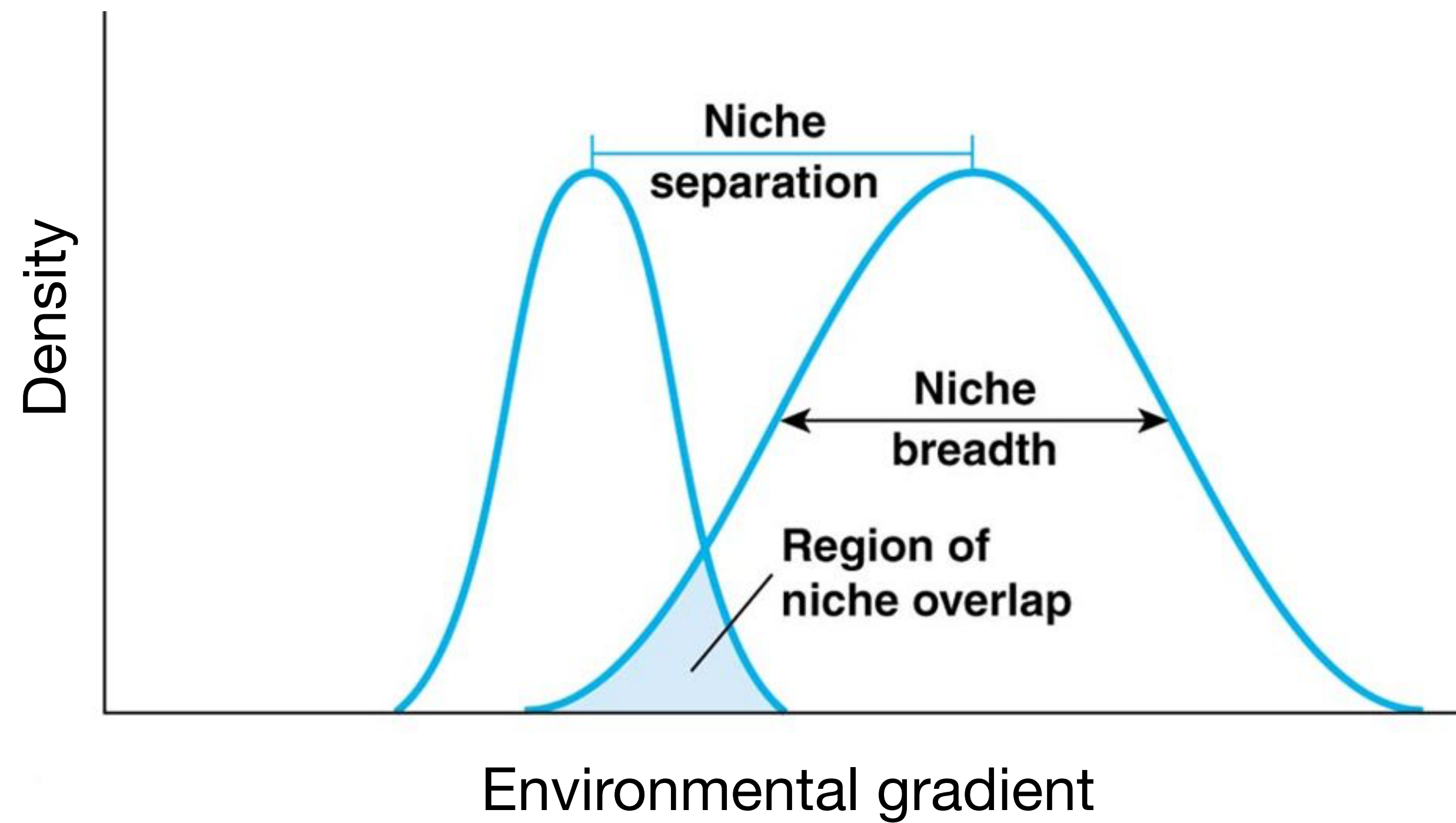


Niche overlap in Species Distribution Modeling

Modeling the distribution of marine biodiversity under global climate change

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Niche overlap analyses

Allow researchers to quantify and compare environmental niches.

Why niche overlap?

Understanding species coexistence and competition: high niche overlap between species, especially in geographic areas where they co-occur, can suggest strong interspecific competition.

Investigating speciation and niche evolution: comparing the niches can provide insights into speciation processes (e.g., sister species).

Did species diverge into different environmental niches (ecological speciation), or do they retain similar niches (niche conservatism), and speciated via other mechanisms like geographic isolation (allopatry)?

Assessing invasion risk: comparing the niche of an invasive species in its native versus invaded range can reveal whether the species has shifted its niche to succeed in the new environment or not (niche conservatism).

Identifying conservation priorities: comparing the niches of different species within a community can help determine if there are ecologically distinct units potentially requiring separate conservation management strategies.

Approaches to measuring niche overlap

Geographic overlap (map comparison);

Response overlap (SDM output functions);

Environmental overlap (environmental space).

Environmental overlap

Schoener's D index is a proportional overlap metric. It tells us how much of the “niche density” or habitat preference overlaps, with more weight given to high-probability areas - places where species are most likely to be.

Ranges from 0 (no overlap) to 1 (identical niches);

Hellinger's I , on the other hand, transforms the probabilities using a square root, which reduces the influence of extreme values (like very high suitability).

Ranges from 0 (no overlap) to 1 (identical niches);

Environmental overlap

Equivalency test: are the species using the same niche?

Similarity test: are the species' niches more or less alike than expected by chance?

If $p < 0.05$, the difference (or similarity) is unlikely due to chance

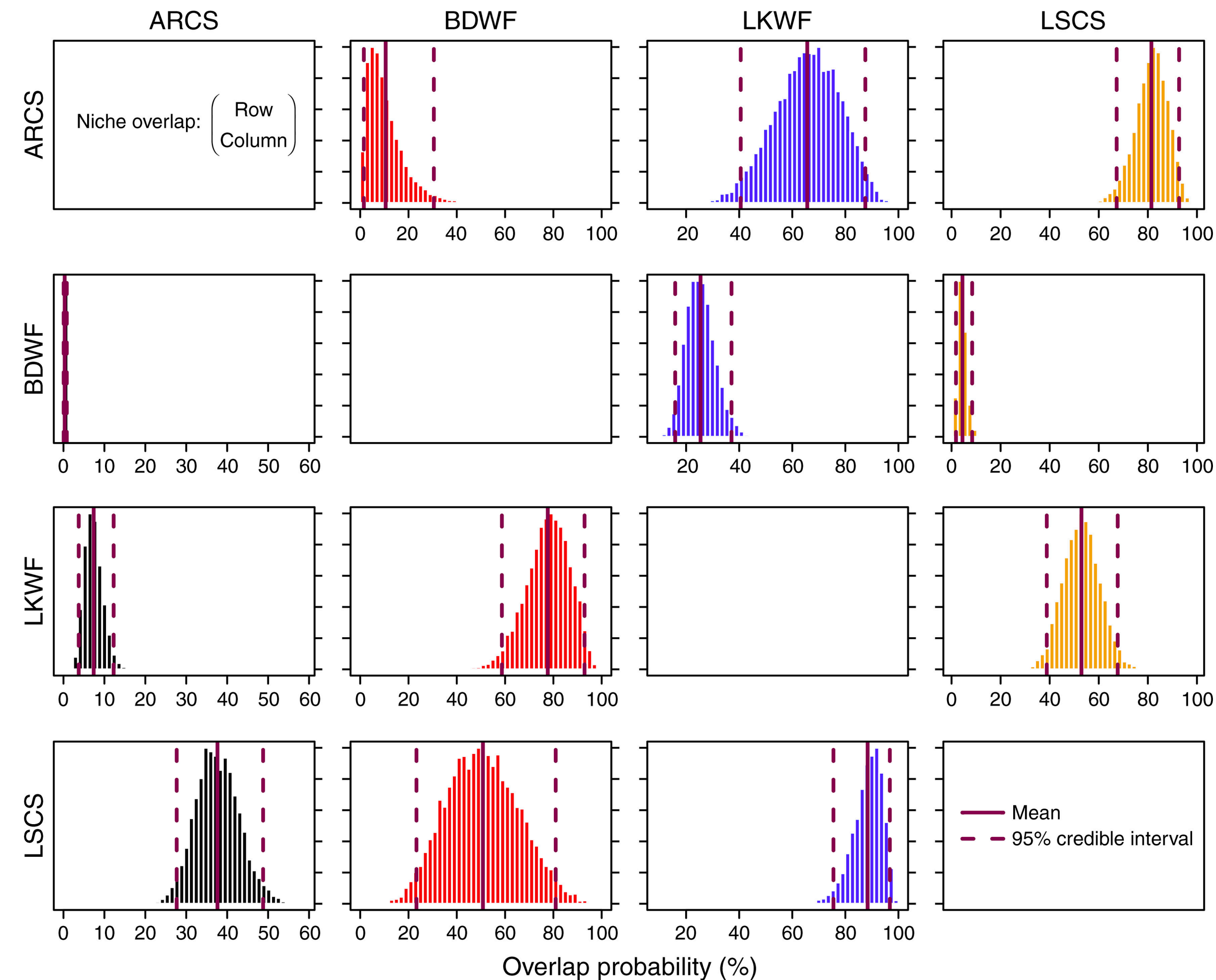
If $p > 0.05$, the observed overlap is within the range of random expectation

Probabilistic method for quantifying n-dimensional niches

Defines overlap as the probability that individuals from species A occur in B's region;

Directional, multivariate, Bayesian framework;

Consistent with Hutchinson's n-dimensional hypervolume;



Posterior distribution of the probabilistic niche overlap metric (%) for a specified niche region - **probability of species displayed in rows overlapping onto those displayed in columns**. Posterior means and 95% intervals in purple.

References

Swanson et al., 2015: <https://doi.org/10.1890/14-0235.1>

Schoener, 1968: <https://doi.org/10.2307/1935534>

Warren et al., 2018: <https://doi.org/10.1111/j.1558-5646.2008.00482.x>